

HEALTHCARE ANALYTICS

Exploratory Data Analysis

BRFSS 2015 Heart Disease Health Indicators Dataset

229,781 respondent records • 45 columns • 10.3% heart disease prevalence

What This Analysis Covers

01

Dataset Snapshot

Scale, target balance, and what the cleaned data contains

02

Univariate Analysis

Individual variable distributions: age, BMI, symptom days, conditions

03

Bivariate Analysis

How risk factors, demographics, and comorbidities relate to heart disease

04

Key Takeaways

The patterns that matter most for population health strategy

A Large, Clean, Imbalanced Survey Dataset

229,781

Cleaned respondent
records

45

Columns, including
12 derived features

10.3%

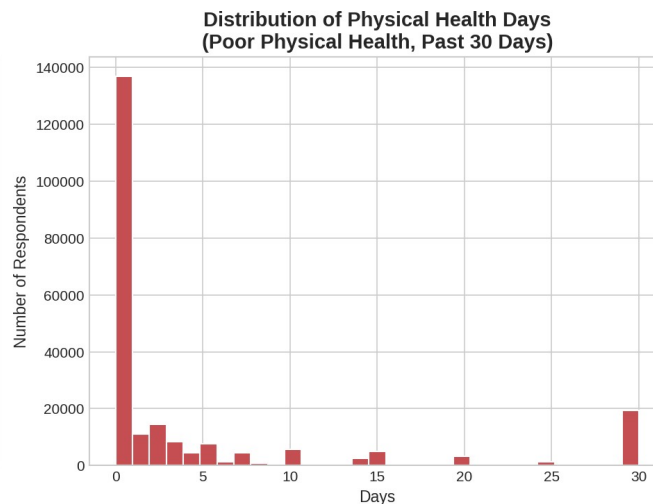
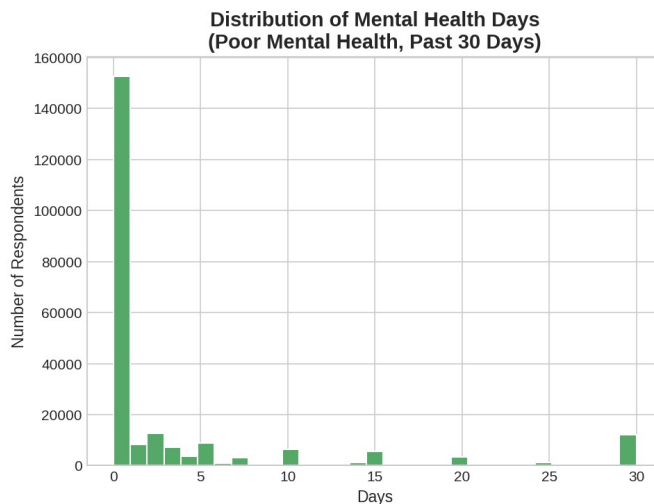
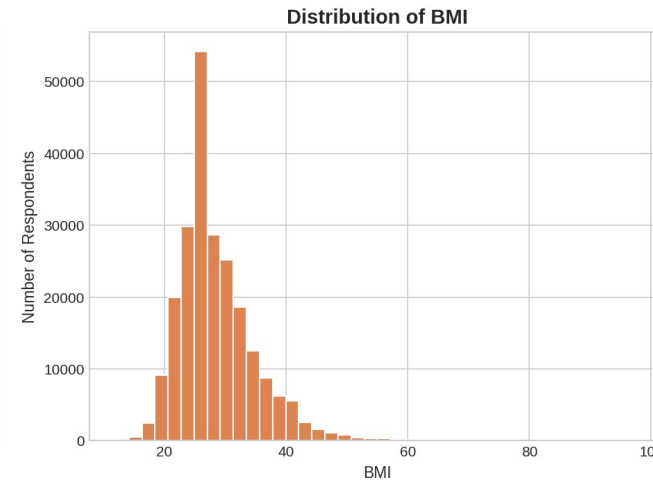
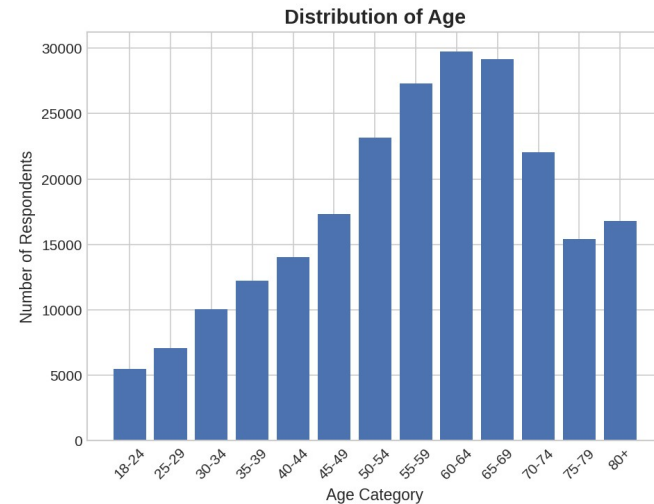
Heart disease /
attack prevalence

9 : 1

Class imbalance
(No : Yes)

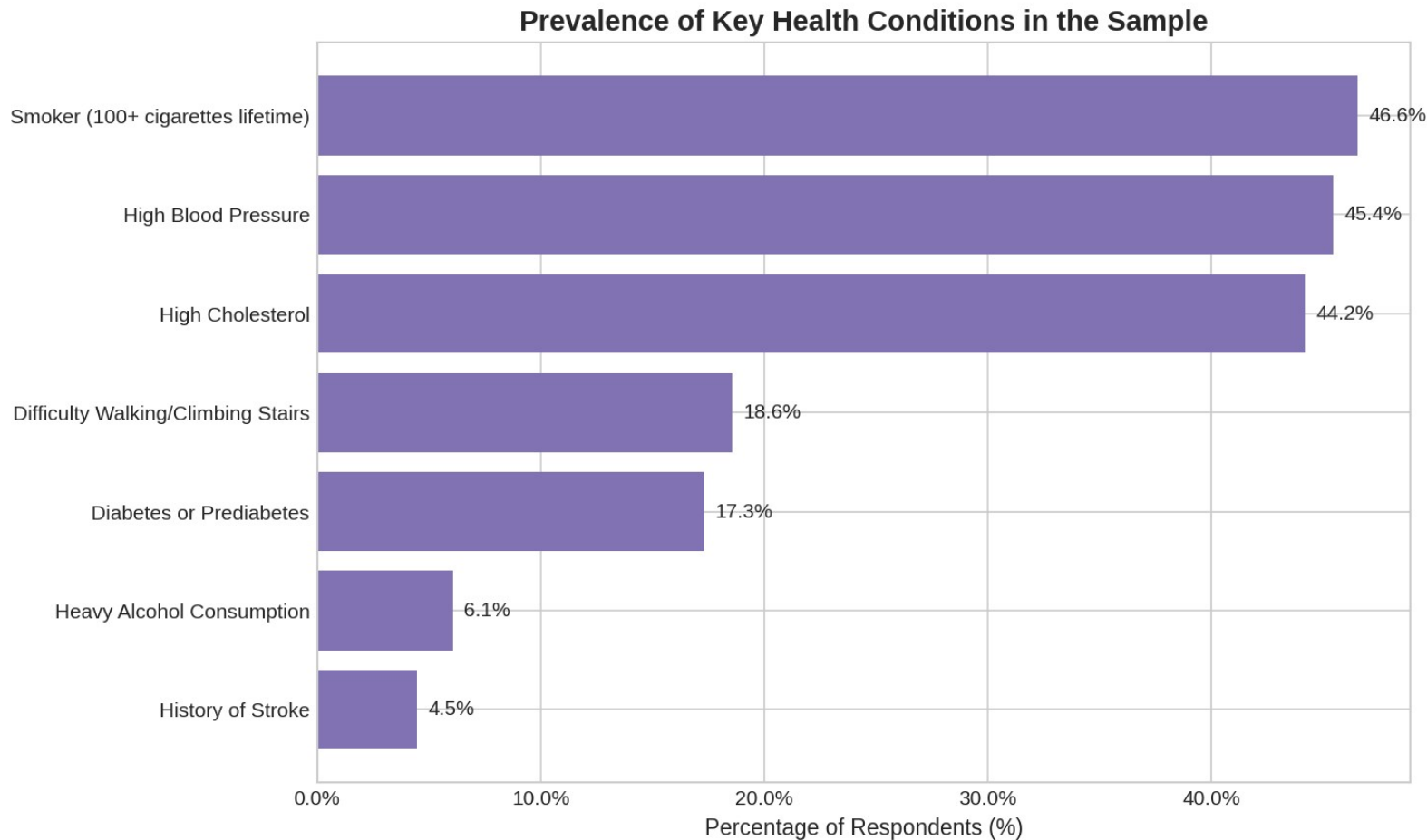
The dataset was deduplicated from 253,680 raw survey rows, then labeled with human-readable categories and 12 derived features (BMI category, broad age group, comorbidity score, and similar) to support this analysis.

Age, BMI, and Self-Reported Health Days



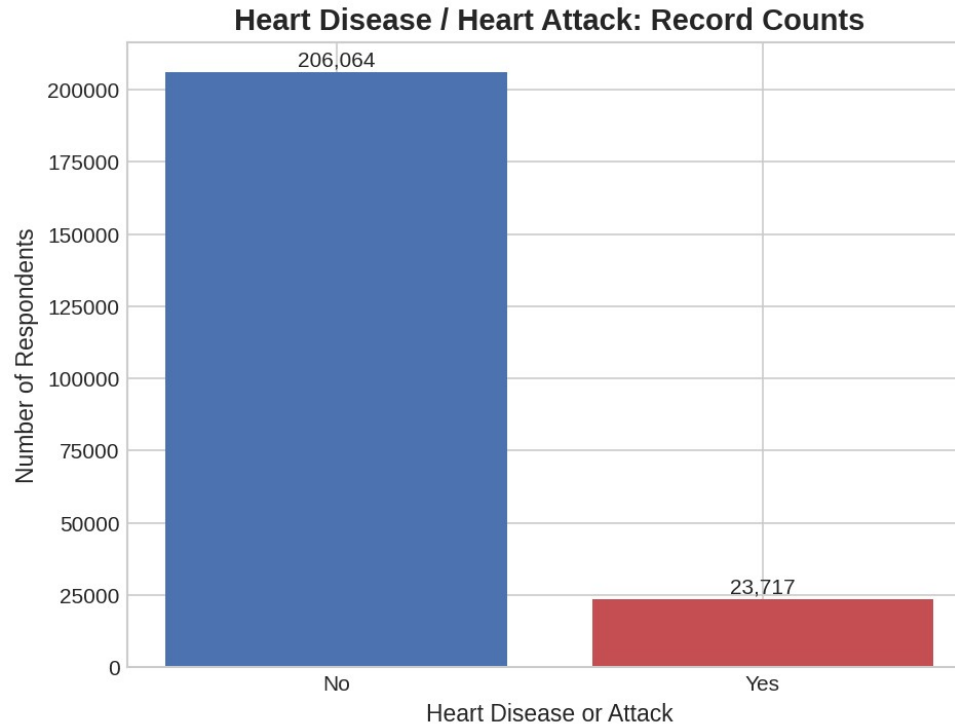
- Respondents skew older — the largest single group falls in the 60–64 age bracket.
- Median BMI is 27.0 (overweight range); the distribution is right-skewed.
- 66% report zero poor mental-health days in the past month.
- 60% report zero poor physical-health days; both variables have a long right tail.

Prevalence of Key Health Conditions

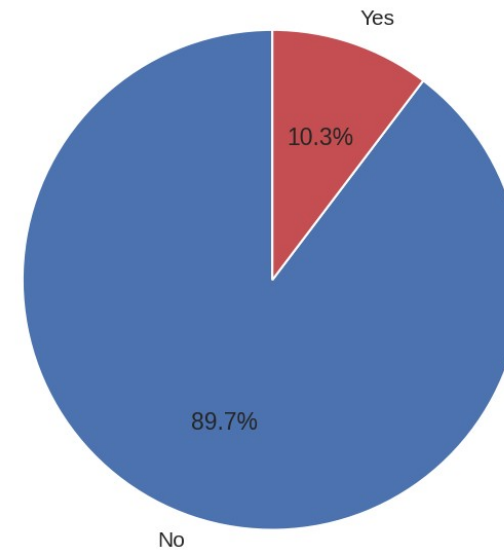


- High blood pressure (45.4%) and high cholesterol (44.2%) are the most common risk factors in the sample.
- Nearly half of respondents (46.6%) report a smoking history of 100+ cigarettes.
- Stroke history (4.5%) and heavy alcohol use (6.1%) are comparatively rare.
- 17.3% report diabetes or prediabetes.

Heart Disease: A Minority, Imbalanced Outcome

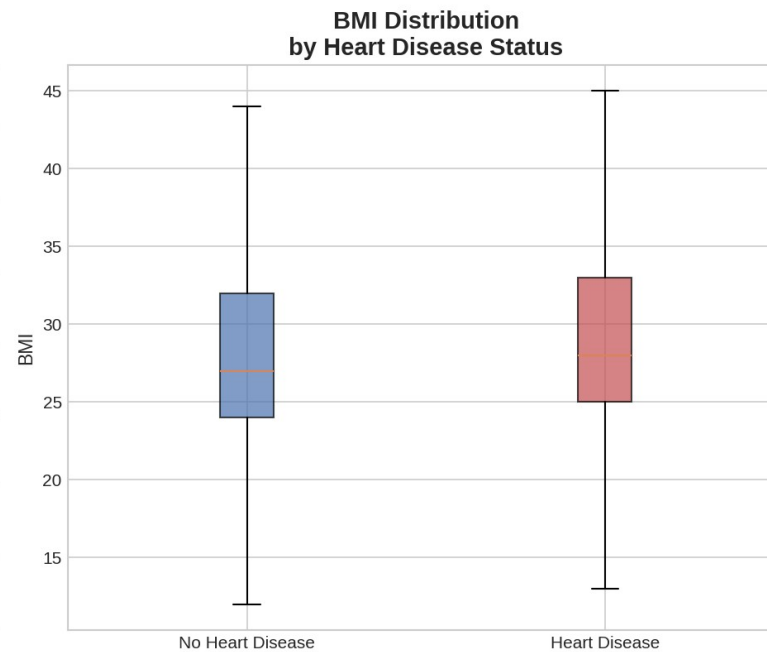
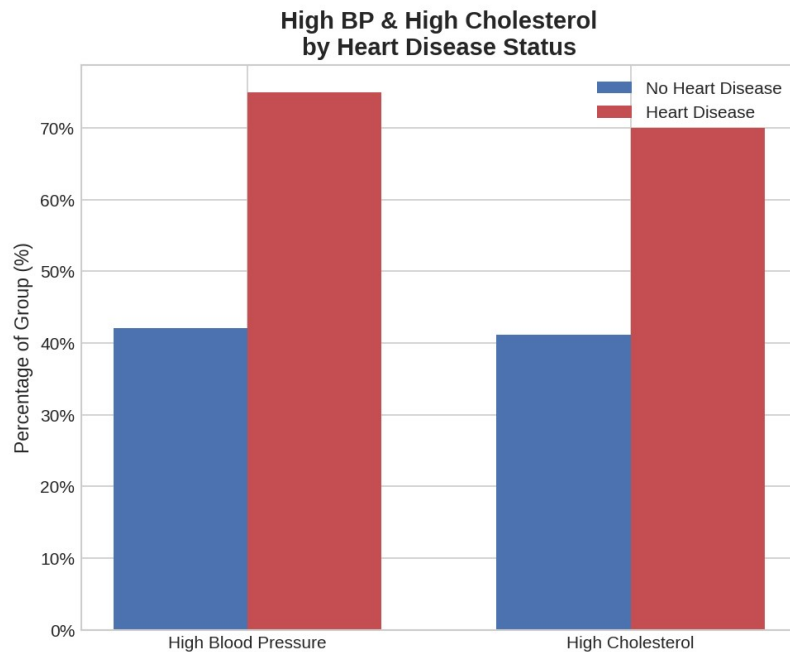


Heart Disease / Heart Attack: Share of Sample



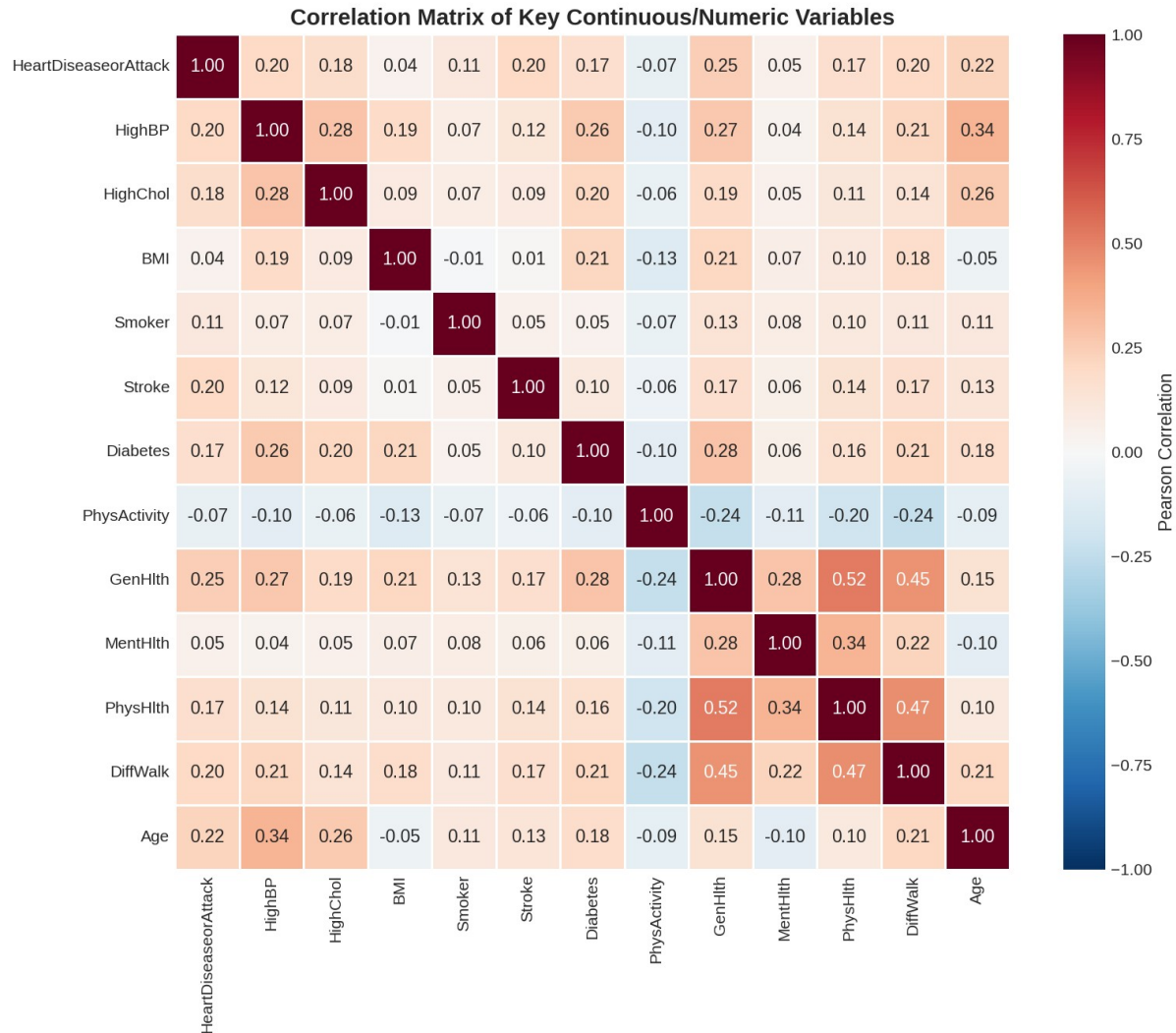
206,064 respondents report no heart disease or attack, versus 23,717 who do — a roughly 9-to-1 split. This imbalance matters: it should inform how any future predictive model is trained and evaluated (e.g., class weighting, resampling, or threshold tuning), since raw accuracy would be a misleading metric on its own.

Known Risk Factors Hold Up in the Data



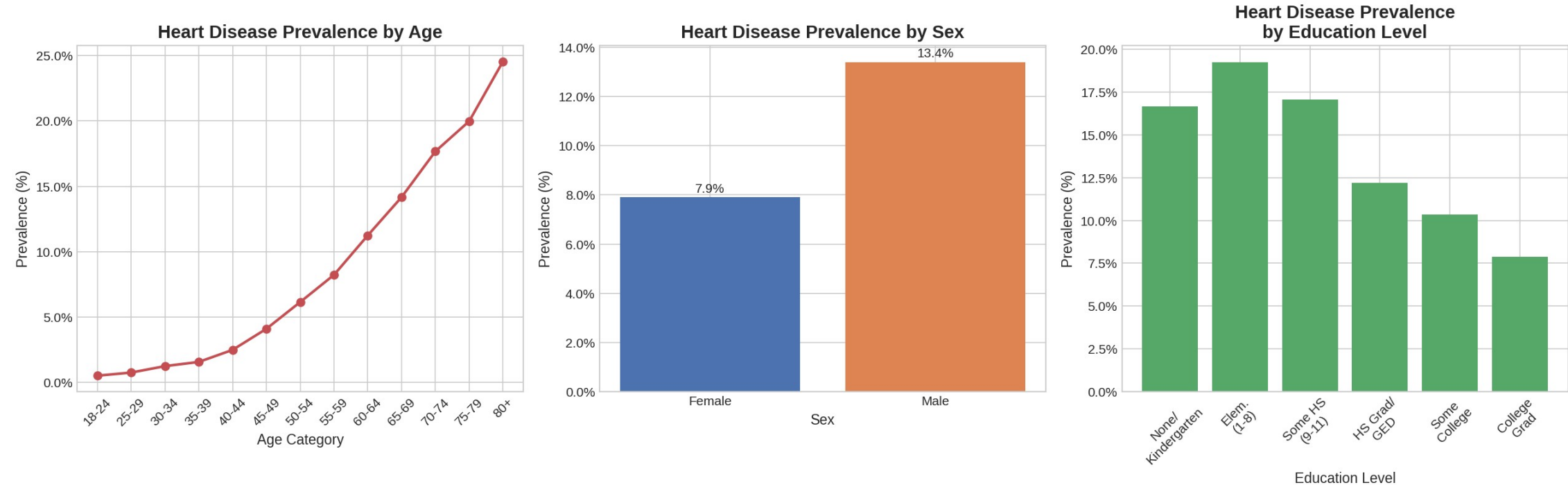
- 75.0% of respondents with heart disease have high blood pressure, vs. 42.0% of those without.
- 70.0% of respondents with heart disease have high cholesterol, vs. 41.2% of those without.
- Median BMI is only slightly higher in the heart disease group (28.0 vs. 27.0) — the distributions overlap substantially.

No Single Variable Dominates



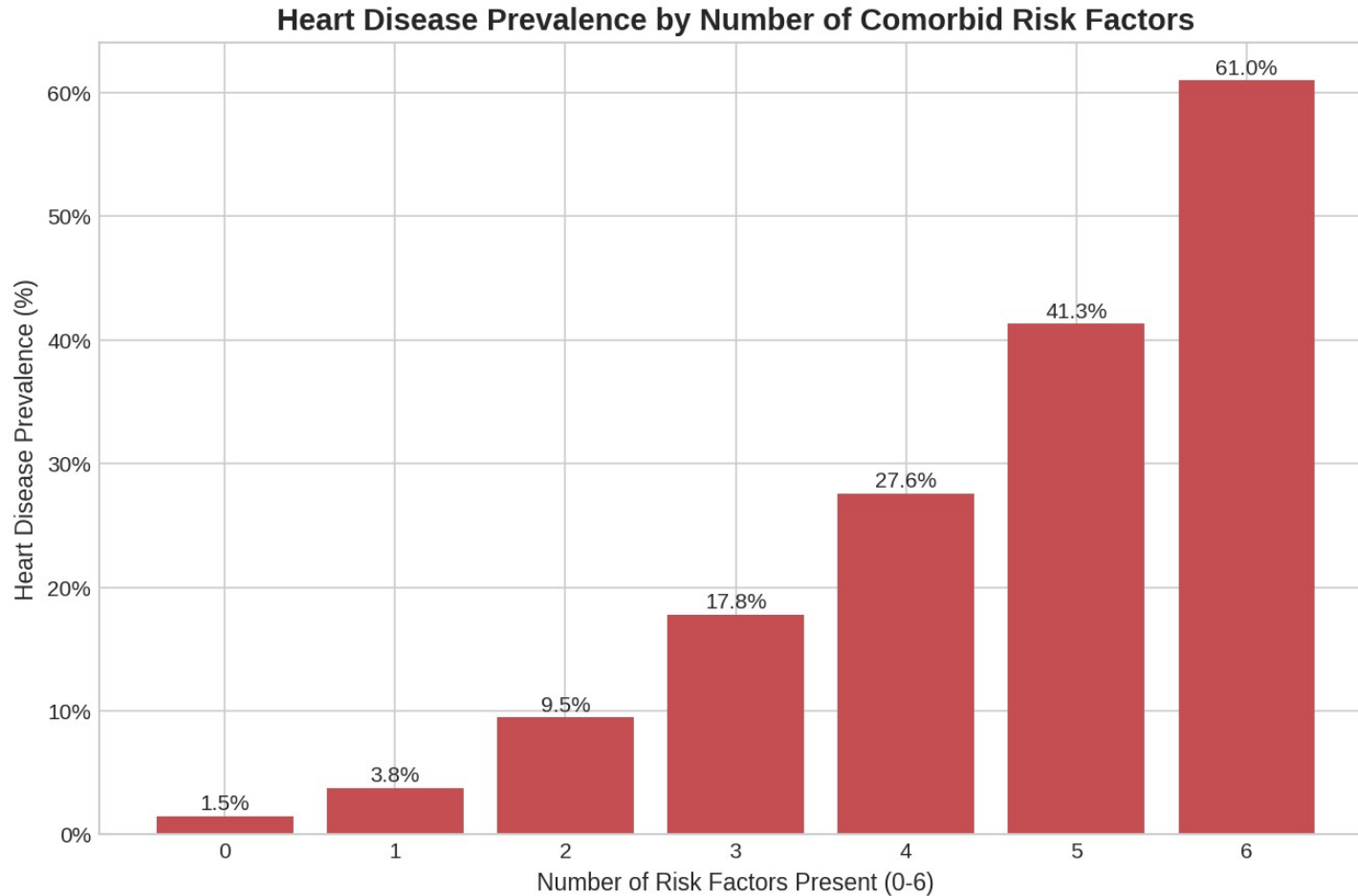
- General health rating shows the strongest correlation with heart disease ($r = 0.25$), followed by age (0.22) and difficulty walking (0.20).
- Physical activity is negatively correlated ($r = -0.07$), consistent with its protective effect.
- All individual correlations are modest (below 0.3) — heart disease risk here is multi-factorial, not driven by one dominant variable.

Age Is the Dominant Demographic Driver



Prevalence climbs steadily from 0.5% (age 18-24) to 24.6% (age 80+). Male respondents show higher prevalence than female respondents (13.4% vs. 7.9%). Prevalence also falls as education rises — from 19.3% among those with only an elementary education to 7.9% among college graduates.

Risk Compounds With Each Added Factor



2.6x

the baseline prevalence rate, for the high-risk cohort (age 60+ with 3+ risk factors, n = 45,327)

- Prevalence rises from 1.5% with zero risk factors to 61.0% with all six present.
- The pattern is a steady, near-linear dose-response, not a sharp threshold effect.

Key Takeaways

Imbalanced target

Heart disease is present in ~1 in 10 respondents — plan for class imbalance in any downstream modeling.

Age drives risk

Prevalence rises from 0.5% to 24.6% across the age range — the strongest single demographic pattern.

Clinical risk factors confirmed

High blood pressure, high cholesterol, poor general health, and difficulty walking are all elevated among heart disease cases.

Risk compounds

Prevalence climbs steadily with each additional comorbid risk factor; a high-risk profile shows 2.6x the baseline rate.

No single predictor dominates

All individual correlations are modest — heart disease risk here reflects a combination of factors, not one variable.

Thank You

Healthcare Analytics — BRFSS 2015 Heart Disease Health Indicators — Exploratory Data Analysis